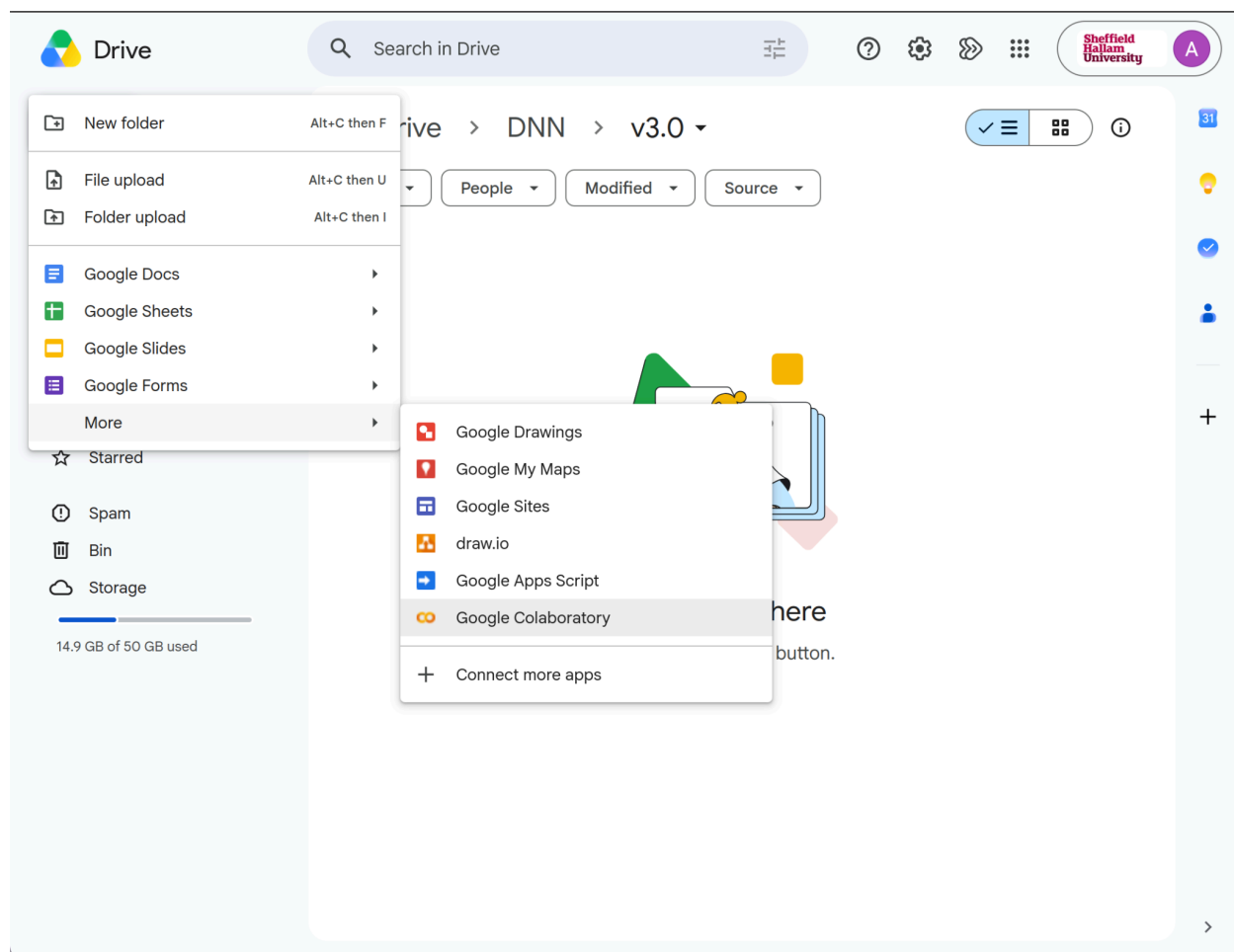




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https://github.com/aljiro/dnnls_lecture_notes_exercises



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Repository:

Branch:

aljiro/
dnnls_lecture_notes_exercises

main

Path



[assessment/assessment_investigation.ipynb](#)



[notebooks/week10_investigations.ipynb](#)



[notebooks/week11_investigations.ipynb](#)



[notebooks/week1_investigations.ipynb](#)



Cancel



week1_investigations.ipynb

File Edit View Insert Runtime Tools Help

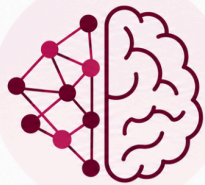


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Deep Neural Networks and Learning Systems

Week 1 - Experiments

Follow the instructions for each experiment in the repository. For every prompt, copy the indicated text into Gemini and use its response to carry out the investigation in this notebook.

Experiment repository: [dnnls_lecture_notes_exercises](#)

Toggle Gemini

Variables Terminal



The screenshot shows a Google Colab interface for a notebook named 'week1_investigations.ipynb'. The top bar includes the Colab logo, a 'PRO' badge, and a menu with 'File', 'Edit', 'View', 'Insert', 'Runtime', 'Tools', and 'Help'. On the right, there are 'Share' and 'Connect' buttons. Below the top bar, a toolbar shows 'Commands', '+ Code', '+ Text', 'Run all', and 'Copy to Drive'. A left sidebar contains icons for file management and execution. The main content area features a background image with the text 'Deep Neural Networks' and the 'Sheffield Hallam University' logo. A white dialog box titled 'Review Custom Instructions' is centered on the screen. It contains the text: 'This shared notebook includes custom instructions that affect how Gemini behaves. Please review the instructions before starting as they were created by someone else.' Below this text is a blue link 'View Custom Instructions'. At the bottom of the dialog are two buttons: 'Use default' and 'Use Custom Instruction'. Below the dialog, the notebook content is partially visible, showing the heading 'Week 1 - Ex' and a paragraph: 'Follow the instructions for each experiment in the repository. For every prompt, copy the indicated text into Gemini and use its response to carry out the investigation in this notebook.' Below this is a link: 'Experiment repository: [dnnls_lecture_notes_exercises](#)'. At the bottom of the Colab interface, there are tabs for 'Variables' and 'Terminal', and a blue star icon.

week1_investigations.ipynb

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Week 1 - Ex

Use default Use Custom Instruction

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[View Custom Instructions](#)

Follow the instructions for each experiment in the repository. For every prompt, copy the indicated text into Gemini and use its response to carry out the investigation in this notebook.

Experiment repository: [dnnls_lecture_notes_exercises](#)

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Deep Neural Networks and Learning Systems

Week 1 - Experiments

Follow the instructions for each experiment in the repository. For every prompt, copy the indicated text into Gemini and use its response to carry out the investigation in this notebook.

Experiment repository: [dnnls_lecture_notes_exercises](#)

Gemini

⌵ ⋮ ✕

Hello, Alejandro

How can I help you today?

How can I install Python libraries?

Load data from Google Drive

Show an example of training a simple ML model

Default

Learn Mode

Visualization Mode

✓ Week1 investigations

Whi Configure

+ Gemini 2.5 Flash ▶

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New notebook in Drive

Open notebook Ctrl+O

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Upload notebook

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Save a copy in Drive

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Follow the instructions for each experiment in the repository. For every prompt, copy the indicated text into Gemini and use its response to carry out the investigation in this notebook.

Experiment repository: [dnnls_lecture_notes_exercises](#)

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04-model-learning-engineering

05-convolutional-inductive-bias

06-explainability-interpretability

07-language-sequence-models

08-generative-modelling

09-transformers-information-ro...

10-scaling-representation-geom...

11-learning-during-inference

.gitignore

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aljiro

Trim Week 11 Gemini instructions below limit7b6cc2e · 2 hours agoHistory

Name	Last commit message	Last commit da...
..		
01-fundamentals	Remove old distributions experiment n...	8 hours ago
02-linear-projections	Allow brief Markdown prefaces within i...	5 hours ago
03-mlp-projections	Allow brief Markdown prefaces within i...	5 hours ago
04-model-learning-engineering	Add Week 4 Experiment 5 diagnostic ca...	4 hours ago
05-convolutional-inductive-bias	Trim Week 5 Gemini instructions below...	4 hours ago
06-explainability-interpretability	Trim Week 6 Gemini instructions below...	3 hours ago
07-language-sequence-models	Restore Week 7 Experiment 3 on autor...	3 hours ago
08-generative-modelling	Trim Week 8 Gemini instructions below...	3 hours ago
09-transformers-information-ro...	Trim Week 9 Gemini instructions below...	3 hours ago
10-scaling-representation-geom...	Trim Week 10 Gemini instructions belo...	2 hours ago
11-learning-during-inference	Trim Week 11 Gemini instructions belo...	2 hours ago

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> 07-language-sequence-models

> 08-generative-modelling

dnnls_lecture_notes_exercises / weeks / 01-fundamentals

aljiro

Remove old distributions experiment numbering

34ed9d1 · 8 hours ago

History

Name	Last commit message	Last commit da...
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experiments	Remove old distributions experiment n...	8 hours ago
notebook-instructions.md	Clarify experiment prompt and investig...	8 hours ago

github.com/aljiro/dnnls_lecture_notes_exercises/tree/main/weeks/01-fundamentals/experiments

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01-sampling.md

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and answer the questions before continuing.

Prompt 1

Copy this into Gemini:

Load the MNIST training dataset with torchvision. Do not train a model. Randomly select 20 images from the training set and show them in a grid with their digit labels. Also print the number of examples of each digit in this sample. Use a fixed random seed so we can reproduce the result.

Questions

- Looking only at these 20 examples, which digits would you think are common in MNIST?
- Which digits would you think are rare?
- Are there any digits missing completely?
- If this were the only data you had seen, what claims about MNIST would you be tempted to make?

Prompt 2

Using the same MNIST dataset and the same random seed, repeat the sampling experiment for sample sizes 20, 50, 100, 500, 1000 and 5000. Do not show the images this time. For each sample size, calculate the proportion of each digit and plot the ten proportions as lines as sample size increases. Also calculate the true digit proportions in the complete training set and show them as horizontal

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Deep Neural Networks
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Week 1 - Experiments

Follow the instructions for each experiment in the repository. For every prompt, copy the indicated text into Gemini and use its response to carry out the investigation in this notebook.

Experiment repository: [dnnls_lecture_notes_exercises](#)

Gemini

⌵ ⋮ ✕

Hello, Alejandro

How can I help you today?

Show me how to start Experiment 1

List the available experiments in the repository

Explain the structure for logging investigations in this notebook

Load the MNIST training dataset with torchvision. Do not train a model. Randomly select 20 images from the training set and show them in a grid with their digit labels. Also print the number of examples of each digit in this sample. Use a fixed random seed so we can reproduce the result.

+ Gemini 2.5 Flash ▶

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 week1_investigations.ipynb [Save in GitHub to keep changes](#)

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▶ Gemini

```
[ ]
import torch
import torchvision
from torchvision import transforms
import matplotlib.pyplot as plt
import random
from collections import Counter

# INVESTIGATION 1
# KEY: Set a fixed random seed for reproducibility
# TRY: Change this seed and observe how the random sample
RANDOM_SEED = 42
torch.manual_seed(RANDOM_SEED)
random.seed(RANDOM_SEED)

# Define a simple transformation to convert images to tensors
transform = transforms.ToTensor()

# Load the MNIST training dataset
# The root='./data' specifies where to download the dataset
mnist_train_dataset = torchvision.datasets.MNIST(
    root='./data',
    train=True,
    download=True,
    transform=transform
)

# KEY: Define how many random images to select
num_samples = 20

# Randomly select indices for the images
# We take a sample of indices from the range of the dataset
sampled_indices = random.sample(range(len(mnist_train_data
```

Gemini

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Load the MNIST training dataset with torchvision. Do not train a model. Randomly select 20 images from the training set and show them in a grid with their digit labels. Also print the number of examples of each digit in this sample. Use a fixed random seed so we can reproduce the result.

⬡ Okay, I'll load the MNIST training dataset, select 20 random images, display them with their labels in a grid, and then print the count of each digit in this sample. I will use a fixed random seed for reproducibility.

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▶ Accept & Run ✓ Accept

✕ Cancel

What can I help you build?

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```
import torch
import torchvision
from torchvision import transforms
import matplotlib.pyplot as plt
import random
from collections import Counter

# INVESTIGATION 1
# KEY:
# TRY:
RANDOM
torch.nn
random

# Define
transform

# Load
# The
mnist_t

    root='./data',
    train=True,
    download=True,
    transform=transform
)

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Gemini

A

Load the MNIST training dataset with torchvision. Do not train a model. Randomly select 20 images from the training set and show them in a grid with their digit labels. Also print the number of digits in this sample. Use a random seed so we can reproduce the results.

What can I help you build?

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Warning: This notebook was not authored by Google

This notebook is being loaded from [GitHub](#). It may request access to your data stored with Google, or read data and credentials from other sessions. Please review the source code before executing this notebook.

Cancel Run anyway

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Terminal

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Save in GitHub to keep changes

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RAM Disk

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```
for digit, count in sorted(digit_count.items()):
    print(f"Digit {digit}: {count}")
```

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100% | 28.9k/28.9k [00:00<00:00, 696kB/s]

100% | 1.65M/1.65M [00:00<00:00, 5.94MB/s]

100% | 4.54k/4.54k [00:00<00:00, 8.88MB/s]

Randomly Selected 20 MNIST Images with Labels

Label: 9

Label: 4

Label: 7

Label: 2

Label: 9

Label: 7

Label: 5

Label: 6

Label: 9

Label: 9

Label: 9

Label: 8

Label: 9

Label: 9

Label: 0

Label: 5

Label: 3

Label: 0

Label: 2

Label: 7

Number of examples of each digit in the sample:
Digit 0: 2
Digit 2: 2
Digit 3: 1
Digit 4: 1
Digit 5: 1
Digit 6: 1
Digit 7: 2
Digit 8: 1
Digit 9: 5

Gemini

A

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Show me how to start Experiment 1

List the available experiments in the repository

What can I help you build?

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2:35 AM Python 3

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